# 燕 山 大 学

# 本科毕业设计（论文）中期检查

# 英文文献阅读与翻译

课题名称: 基于深度图神经网络的板状结构超声损伤定位方法

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| 英文文献 1 | |
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| **文献信息** | Ma J, Zhao S, Yang Z, et al. MCAGN: Multi-layer Coupled Attention Graph Networks for Composites Damage Detection[J]. IEEE Sensors Journal, 2025. |
| 英文题目 | MCAGN: Multilayer Coupled Attention Graph Networks for Composites' Damage Detection |
| 英文摘要 | Damage detection in carbon-fiber-reinforced polymer (CFRP) composites is essential for maintaining structural integrity and preventing catastrophic accidents. Recently, machine-learning-based damage detection methods have garnered increasing attention. However, existing approaches often face challenges in extracting relational information about damage, both within guided wave paths and between damage samples. These limitations result in insufficient detection stability under varying training sample conditions and increase the risk of model overfitting when relying solely on high-dimensional information extracted from guided wave signals. To address these challenges, this article proposes a multilayer graph network with a coupled attention mechanism based on feature fusion. In the proposed method, by encoding sensor coordinates and fusing guided wave signal features, positional information can be integrated into the guided wave signals. A multilayer graph network is further used to extract signal features, while the interlayer coupled attention mechanism and multilayer graph construction fully explore the relational information associated with the damage. Experimental datasets collected from real CFRP composites demonstrate the effectiveness of the proposed method. Comparative results show that the multilayer graph neural network (GNN) achieves superior stability and accuracy, even with limited training samples and noisy training samples. |
| **中文题目** | MCAGN：用于复合材料损伤检测的多层耦合注意力图网络 |
| 中文摘要: | 碳纤维增强聚合物（CFRP）复合材料的损伤检测对保障结构完整性、防止灾难性事故至关重要。近年来，基于机器学习的损伤检测方法日益受到关注。然而，现有方法在提取损伤相关关系信息方面仍面临挑战，既难以刻画导波传播路径内部的损伤关联，也难以建模不同损伤样本之间的关联。这些局限导致模型在训练样本数量或质量变化时检测稳定性不足，且若仅依赖从导波信号中提取的高维特征，则更易发生过拟合。为应对上述问题，本文提出一种基于特征融合的多层图网络，并引入层间耦合注意力机制。该方法通过编码传感器空间坐标并融合导波信号特征，将位置信息嵌入导波信号；再借助多层图网络提取信号特征，同时利用层间耦合注意力机制与多层图构建策略，充分挖掘与损伤相关的各类关系信息。在真实 CFRP 复合材料采集的实验数据集上验证了所提方法的有效性。对比结果表明，所提出的多层图神经网络（GNN）即便在训练样本有限或存在噪声的情况下，仍能保持优异的检测稳定性和精度。 |
| 文献理解 | 该文献创新的将多层图结构与层间耦合注意力机制相结合，既建模单个损伤内各导波路径间的内部关联，又挖掘不同损伤样本间的外部共性；同时引入高斯坐标映射实现传感器位置信息与信号特征的深度融合，有效缓解高维特征导致的过拟合，显著提升小样本与噪声下的泛化能力与鲁棒性。 |

| 英文文献 2 | |
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| **文献信息** | Zhao Y, Tang D, Ding C, et al. TFA_Net: A Time-Frequency Attention Network for Enhancing Defect Recognition and Noise Robustness in Ultrasonic Guided Wave Pipeline Inspection[J]. International Journal of Pressure Vessels and Piping, 2025: 105735. |
| 英文题目 | TFA_Net: A time-frequency attention network for enhancing defect recognition and noise robustness in ultrasonic guided wave pipeline inspection |
| 英文摘要 | Ultrasonic guided wave-based pipeline defect detection is crucial for achieving efficient and cost-effective structural health monitoring of pipelines. However, signal attenuation and noise interference during wave propagation severely hinder the accurate identification of defect signals. To address this challenge, a Time- Frequency Attention Network (TFA_Net) is proposed, integrating time-frequency domain signal processing with deep learning techniques to improve defect recognition accuracy and enhance noise robustness in ultrasonic guided wave signals. First, the Ultrasonic Guided Wave Defect Dataset (UGW-Dataset) was established, including experimental and simulated data, covering cracks and corrosion defects of various sizes and shapes. Next, a time- frequency attention block (TFA_Block) was designed in TFA_Net to perform multi-scale feature extraction in both time and frequency domains, enabling effective capture of global and local characteristics of defect signals. Experimental results demonstrate that TFA_Net achieves a defect recognition accuracy of 99.4 % on the UGW-Dataset, confirming its exceptional feature extraction and defect recognition capabilities. Furthermore, TFA_-Block significantly enhances the robustness of TFA_Net within the signal-to-noise ratio (SNR) range from 40 dB to 10 dB, effectively mitigating the negative impact of noise on recognition accuracy. This study provides an efficient and noise-resilient approach for defect recognition in pipeline structural health monitoring (SHM). |
| **中文题目** | TFA_Net：一种面向超声导波管道检测的时频注意力网络，用于提升缺陷识别精度与抗噪能力 |
| 中文摘要: | 基于超声导波的管道缺陷检测对实现高效、低成本的管道结构健康监测至关重要。然而，导波传播过程中的信号衰减与噪声干扰严重阻碍了缺陷信号的准确识别。为应对这一挑战，本文提出一种时频注意力网络（TFA_Net），将时频域信号处理与深度学习技术相结合，以提升缺陷识别精度并增强超声导波信号的抗噪能力。首先，构建了超声导波缺陷数据集（UGW-Dataset），涵盖实验与仿真数据，覆盖多种尺寸与形状的裂纹与腐蚀缺陷；其次，在 TFA_Net 中设计了时频注意力模块（TFA_Block），在时域与频域同步执行多尺度特征提取，从而有效捕获缺陷信号的全局与局部特征。实验结果表明，TFA_Net 在 UGW-Dataset 上实现了 99.4%的缺陷识别准确率，验证了其卓越的特征提取与缺陷识别能力；此外，TFA_Block 显著提升了 TFA_Net 在信噪比（SNR）为 40 dB 至 10 dB 范围内的鲁棒性，有效缓解了噪声对识别精度的负面影响。本研究为管道结构健康监测（SHM）中的缺陷识别提供了一种高效且抗噪的解决方案。 |

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| 文献理解 | 该文献将时频注意力模块和频域增强机制融合在 CNN-Transformer 混合架构中，频域增强机制通过 FFT-可学习频带选择-IFTT 实现物理意义明确的频域注意力，GW-MSA 提升全局建模效率，显著增强导波信号在噪声下对裂纹、腐蚀等缺陷的判别能力。 |

| 英文文献 3 | |
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| **文献信息** | Sun L, Song R, Wei J, et al. Physics-Augmented Spatial-Temporal graph convolutional network for damage localization using Ultrasonic guided waves[J]. Mechanical Systems and Signal Processing, 2024, 221: 111738. |
| 英文题目 | Physics-Augmented Spatial-Temporal graph convolutional network for damage localization using Ultrasonic guided waves |
| 英文摘要 | Structural health monitoring technology based on ultrasonic guided waves (UGW) is a potential application for damage detection in plate-like composite structures. This paper introduces a novel physics-augmented spatial–temporal graph convolutional network (P-STGCN) for the damage detection task in composite plate. Firstly, the graph topology containing edge weight attributes is constructed based on the spatial correlation of the sensors and the UGW scattering principle, which strengthens the expression of the data spatial information of the data. Then, a spatial feature extractor and a temporal feature extractor are constructed based on graph convolutional network and convolutional neural network, respectively. Further, the spatial–temporal fusion method is proposed for the first time in the field of UGW damage detection, which enables the perception of multi-level damage information. In addition, a sample compensation attention module based on wavefield scattering energy is designed to compensate for the samples' spatial distribution. Finally, a dataset was constructed in a composite plate containing only four sensors to evaluate the model's performance. The results show that the average localization error is 10.36 mm. Compared with other deep learning localization methods, the P-STGCN model is superior in both localization accuracy and computational resources. |
| **中文题目** | 基于超声导波的物理增强型时空图卷积网络用于损伤定位 |
| 中文摘要: | 基于超声导波（UGW）的结构健康监测技术在板状复合结构损伤检测中具有潜在应用价值。本文针对复合板中的损伤检测任务，提出一种新型物理增强型时空图卷积网络（P-STGCN）。首先，基于传感器的空间相关性和 UGW 散射原理构建包含边权重属性的图拓扑结构，强化了数据空间信息的表达。接着，分别基于图卷积网络和卷积神经网络构建空间特征提取器和时间特征提取器。进一步地，首次在 UGW 损伤检测领域提出时空融合方法，实现了多层次损伤信息的感知。此外，设计了一种基于波场散射能量的样本补偿注意力模块，用于补偿样本的空间分布。最后，构建了一个仅含四个传感器的复合板数据集以评估模型性能。结果表明，平均定位误差为 10.36 mm。与其他深度学习定位方法相比，P-STGCN 模型在定位精度和计算资源消耗方面均表现更优。 |
| 文献理解 | 本文献将物理先验嵌入时空图卷积网络，提出物理增强型图结构，依据 UGW 散射能量动态构建加权邻接矩阵，设计样本补偿注意力模块，融合波场散射能量实现空间分布自适应加权；并开创性地在 UGW 损伤定位中引入时空特征协同提取与融合机制，显著提升小传感器配置下的定位精度与可解释性。 |

| 英文文献 4 | |
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| **文献信息** | Xu X, Liu C. Physics-guided deep learning for damage detection in CFRP composite structures[J]. Composite Structures, 2024, 331: 117889. |
| 英文题目 | Physics-guided deep learning for damage detection in CFRP composite structures |
| 英文摘要 | Structural health monitoring of carbon fiber reinforced plastics composite structures based on deep learning usually builds on black box models for damage detection. However, the lack of physics knowledge limits its application into real-world problems where structural property or environmental condition varies. To overcome this limitation, we propose a physics-guided deep learning framework to integrate physics into data-driven models. This physics-guided convolutional neural network leverages structural degradation trend and physical consistency by combining the output of the physical model with the observed feature in a hybrid model. This hybrid model uses an additional branch to observe the information of stiffness degradation, which is the input into the physical model to describe the damage growth in structures by establishing a relationship with the power spectral density change in the guided wave signals. Additionally, the physics-based loss function is designed as part of the learning objective to ensure the model outputs satisfy the existing physics and maintain physical consistency. Experiment results show that our method demonstrates great generalizability that by training on data of only one particular CFRP composite structure, the PGCNN model can expand its great performance on all other CFRP layups in the NASA-published CFRP Dataset with over 90% accuracy. |
| **中文题目** | CFRP 复合结构损伤检测中的物理引导深度学习 |
| 中文摘要: | 基于深度学习的碳纤维增强塑料复合结构健康监测通常依赖黑箱模型进行损伤检测。然而，缺乏物理知识限制了其在结构属性或环境条件变化的真实问题中的应用。为克服这一局限，我们提出一种物理引导的深度学习框架，将物理知识融入数据驱动模型。该物理引导卷积神经网络通过在混合模型中结合物理模型输出与观测特征，利用结构退化趋势和物理一致性。该混合模型使用额外分支观测刚度退化信息，作为物理模型的输入，通过建立其与导波信号中功率谱密度变化的关系来描述结构中损伤的增长。此外，设计基于物理的损失函数作为学习目标的一部分，确保模型输出符合现有物理规律并保持物理一致性。实验结果表明，我们的方法展现出强大的泛化能力：仅在一个特定 CFRP 复合结构的数据上训练，PGCNN 模型即可在 NASA 发布的 CFRP 数据集中所有其他 CFRP 铺层上均实现超过 90%的准确率。 |
| 文献理解 | 本本文献将 CFRP 刚度退化趋势与导波功率谱密度变化的物理关系，通过双分支网络架构和基于物理的损失函数深度嵌入深度学习框架，实现端到端的物理引导训练，在仅用单一铺层数据训练下即获得强跨结构泛化能力，显著提升模型可解释性与工程适用性。 |

| 英文文献 5 | |
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| **文献信息** | Liu C, Chen Y, Xu X, et al. Domain generalization-based damage detection of composite structures powered by structural digital twin[J]. Composites Science and Technology, 2024, 258: 110908. |
| 英文题目 | Domain generalization-based damage detection of composite structures powered by structural digital twin |
| 英文摘要 | This research addresses the challenge of generalizing deep learning models for different CFRP composite structures in the task of fatigue damage detection. To overcome this challenge, knowledge distillation is employed to enhance the generalizability of deep learning models. A teacher network processes continuous wavelet transform images using Fourier transform and neural networks, while a student network distills the teacher network. This framework improves the models' generalization performance by transferring knowledge from the teacher network to the student network. Additionally, soft gradient boosting is utilized to further enhance the generalizability. By constructing a main sub-network and multiple parallel auxiliary sub-networks within the teacher network, the student network mimics the main sub-network to achieve improved accuracy in the target domain and prevent overfitting. To augment limited datasets of real CFRP monitoring signals and help to learn domain-invariant features, structural digital twin technology is leveraged to generate simulated monitoring signals, which enables the models to capture domain invariant information, significantly enhancing its performance of fatigue damage detection across different structures. Damage detection based on the gener-alization results between multiple Layups demonstrates a test accuracy exceeding 80 % when the monitoring data of the target CFRP structure is unavailable during training. Therefore, the cross-structure damage detection ability of the proposed approach is well proved. |
| **中文题目** | 基于结构数字孪生的复合结构域泛化损伤检测 |
| 中文摘要: | 本研究针对在不同 CFRP 复合结构之间进行疲劳损伤检测时深度学习模型泛化能力的挑战。为克服这一挑战，采用知识蒸馏来增强深度学习模型的泛化能力。教师网络使用傅里叶变换和神经网络处理连续小波变换图像，学生网络则从教师网络中蒸馏知识。该框架通过将知识从教师网络传递给学生网络，提升了模型的泛化性能。此外，利用软梯度提升进一步增强泛化能力。通过在教师网络中构建主子网络和多个并行辅助子网络，学生网络模仿主子网络，从而在目标域中提高准确性并防止过拟合。为了扩充有限的真实 CFRP 监测信号数据集并帮助学习域不变特征，利用结构数字孪生技术生成模拟监测信号，使模型能够捕捉域不变信息，显著提升在不同结构间的疲劳损伤检测性能。基于多铺层间泛化结果的损伤检测表明，在训练过程中目标 CFRP 结构的监测数据不可用时，测试准确率仍超过 80%。因此，所提方法的跨结构损伤检测能力得到了充分验证。 |
| 文献理解 | 该文献将结构数字孪生与域泛化深度融合，构建了基于知识蒸馏与集成学习的 DGDT 框架；利用 FEM 驱动数字孪生生成高保真模拟信号，弥补真实数据稀缺，强化域不变特征学习，并通过傅里叶相位约束与软梯度提升机 |

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| | 制，显著提升模型跨 CFRP 铺层结构的泛化能力与可解释性。。 |